

SERVICE MANUAL

Invivo Corporation
3545 SW 47th Avenue
Gainesville, FL 32608

Document TR0638S

Revision 1

3.0T GEM SMALL ANTERIOR ARRAY

GE Part Number: M7000GY

Invivo Part Number: 9896-031-12161

Manufactured By:



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GE Healthcare

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UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for tekniker, operatøren eller patienten.
WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING (EN)	This service manual is available in English only. - If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services. - Do not attempt to service the equipment unless this service manual has been consulted and is understood. - Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles - Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. - Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. - Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. - Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. - Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. - Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. - Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. - Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. - Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. - Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. - Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. - Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά. - Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης. - Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις. - Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. - Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
ADVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fáanleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. - Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilið. - Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. - Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. - Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. - Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

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BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. • Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. • Neveiciet aprikojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. • Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.
ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. • Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. • Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. • Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. - Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. - Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. - Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. - Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. - Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. - Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. - Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. - Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. - Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. - Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. - Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. - Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. - Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. - Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. - No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. - La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. . - Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. - Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. - Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
OPOZORILO (SL)	Ta servisni priročnik je na voljo samo v angleškem jeziku. - Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod. - Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli. - Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. - Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. - Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. - Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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ATTENTION!

A GE field engineer **MUST** perform installation of this coil. Please contact your field engineer to arrange for installation.

Attempts to perform installation of this coil without proper system configuration will result in system errors.

SECTION 1 – INTRODUCTION

1.1 Product Identification and Shipping List

This is a service manual for the 3.0T GEM Small Anterior Array. This coil is identified by using the label below:

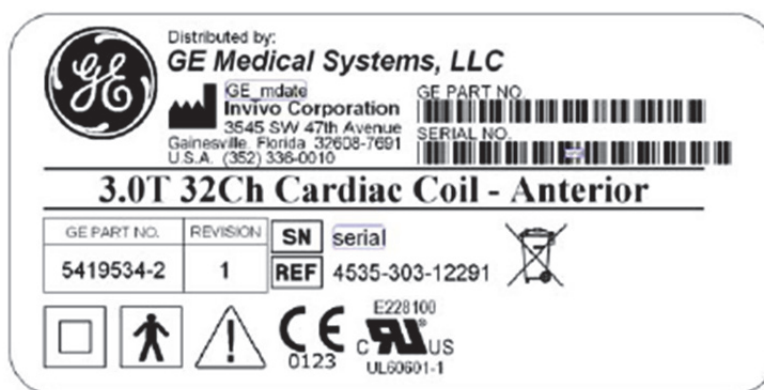


Illustration 1: Anterior Array Label

Illustration 2 shows the 3.0T GEM Small Anterior Array, Phantom Positioner, Anterior Pad, and Strap Set.

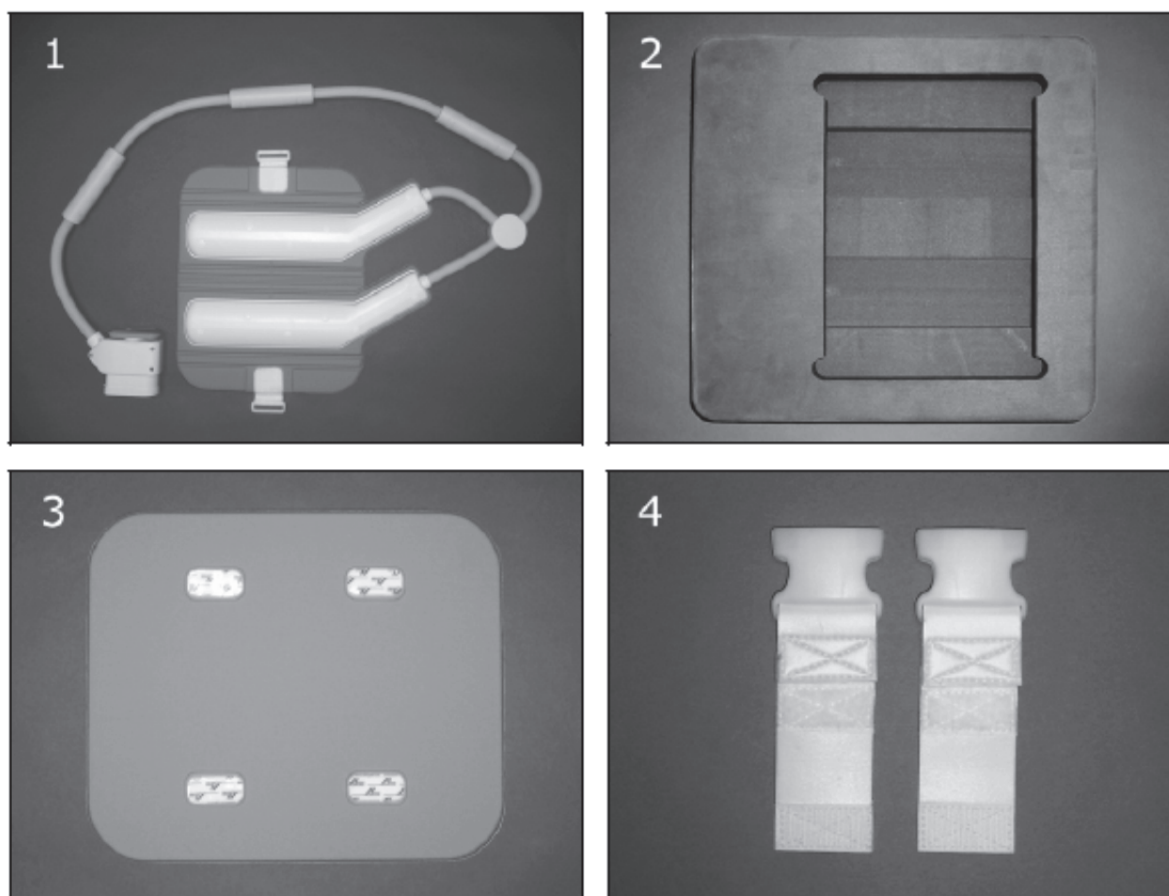


Illustration 2

1.2 Shipping Contents

SHIPPING LIST BOX 1– TABLE 1-2-1

Description	Invivo Part #	GE Part #	Qty
3.0T 32Ch Cardiac Coil - Anterior	4535-303-12291	5419534-2	1
Phantom Positioner, GEM Small Anterior Array (2 pieces)	4535-303-03571	5406404-11	1
Anterior Pad, GEM Small Anterior Array	4535-302-60711	5406404-10	1
Strap Set, GEM Small Anterior Array	4535-303-03591	5406404-12	
Operator Manual	4535-303-05151	N/A	1

1.3 Compatibility

This coil is compatible with the following hardware configurations:

Discovery MR750w Non-GEM*, and Discovery MR750w GEM 3.0T Systems

*In order to use the GEM Small Anterior Array on a MR750w Non-GEM system, the curved phantom positioner part 5366627-4 must be purchased and billed to the customer to run MCQA testing.

1.4 Related Documentation

- 5670000 Discovery MR750w and MR750w GEM 3.0T Service Methods DVD
- 1.5T & 3.0T GEM Small Anterior Array Operator's Manual, 4535-302-58991

1.5 Environmental Requirements

Storage

Store the coil in an air-conditioned scan room or equipment room. To store the coil, a storage space of 48.3 cm/19.0 in (width) x 16.5 cm/6.5 in (depth) x 101.6 cm/40.0 (length) is required.

Dimensions

Coil	48.3 cm x 16.5 cm x 101.6 cm	(19.0 in x 6.5 in x 18.0 in)
Phantom Positioner	50.6 cm x 50.6 cm x 17.7 cm	(20 in x 20 in x 7 in)

Weight

Coil	2.95 Kg	(6.5 lb)
Phantom Positioner	0.68 Kg	(1.5 lb)

SECTION 2: INSTALLATION PROCEDURE

There are no installation steps other than plugging in the coil and performing functional checks (MCQA test). Coil configurations will already be on the system software. The coil required configurations to use this coil are:

COIL CONFIGURATIONS– TABLE 2-1

System	Clinical Coil Name	Annotation Name	Description
MR750w-GEM	GE_32ChCardAntOnly	Small AA only 30	GEM Small AA
	Body 30 Small	Body 30 Small	GEM Chest/Body/Pelvis or GEM Whole Body
MR750w	GE_32Ch Cardiac Ant	32Ch Cardiac Ant	32Ch Cardiac Array by Invivo

These coil configurations can be confirmed present on the system in the “Coil Information” tab of Coil DB Explorer (more information on coil DB Explorer can be found on MR Service Methods under Coils -> Running Coil Database Explorer).

2.1 Installation Finalization

Complete Product Locator information for the coil. Failure to do this may result in failure of your site to receive future FMI's. Instructions are below.

- o Go to GIB: http://gib.gehealthcare.com/gib/gib_MainMenu.jsp
- o After the log in screen, click on [Global Installed Base entries]
- o Under “Serial Changes to Installed Base”, enter the system ID of the system on which this coil is being installed. Click [Submit]
- o Click on [New Entries]

SECTION 3: GEM SMALL ANTERIOR ARRAY MCQA TEST

Follow this procedure to perform the MCQA test. In order to run MCQA for this coil on MR750w Non-GEM, the curved phantom positioner part 5366627-4 must be purchased. For detail, please see the 32Ch Cardiac service manual for MR750w Non-GEM.

3.1 Tools Required

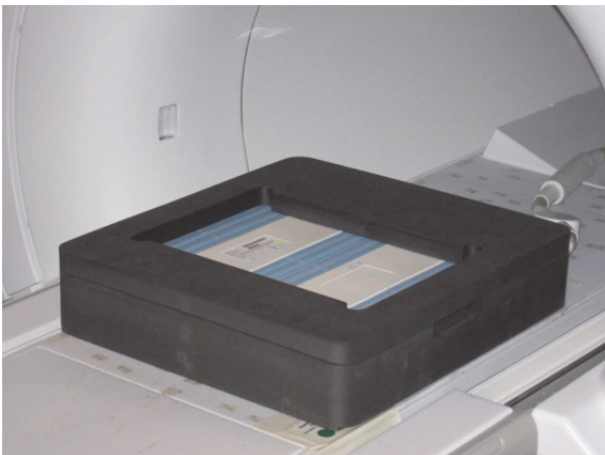
TOOLS REQUIRED - TABLE 3-1

Description	GE Part #	Qty
Unified TL Phantom Brick	5343347	1
GEM Small AA – Phantom Positioner (2 pcs)	5406404-11 (MR750w GEM flat table)	1

3.2 MCQA Setup

All images shown are on the GEM flat table.

TABLE 3-2 Steps for Coil and phantom setup

Visual	Description
	1. Remove any other surface coils (if present) from the cradle.
 Illustration 3  Illustration 4	2. Remove any pads present on the coil. The cable must point towards the magnet and plug into ports P1 or P2. Place lower half of phantom positioner on table. Ensure that the positioner is centered in the R/L direction. Insert coil and place top part of the positioner on the coil as shown in Illustration 3 and Illustration 4.

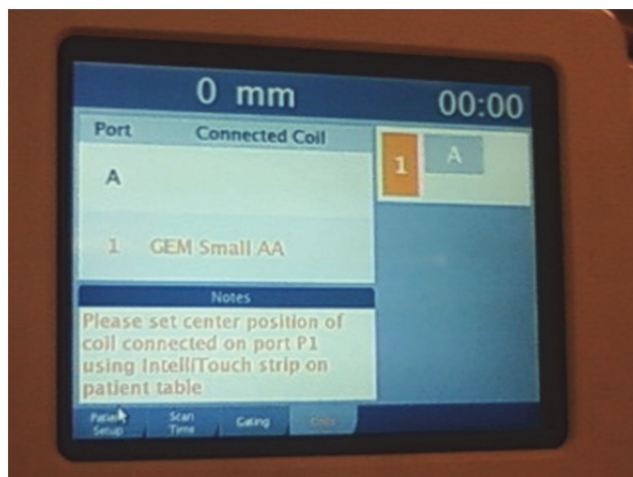


Illustration 5

3. Confirm the message on in-room monitor as shown in Illustration 5.

Use the IntelliTouch strip as shown in Illustration 6.



Illustration 6

Confirm that the message on the in-room monitor has disappeared as shown in Illustration 7.

Push here to use IntelliTouch strip



Illustration 7

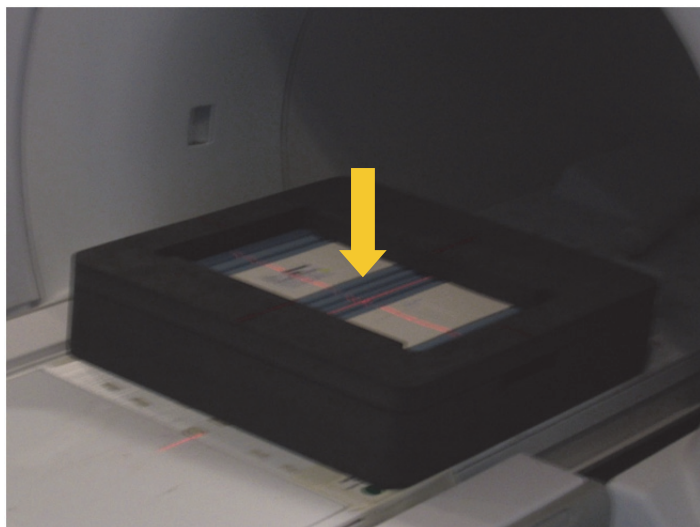


Illustration 8

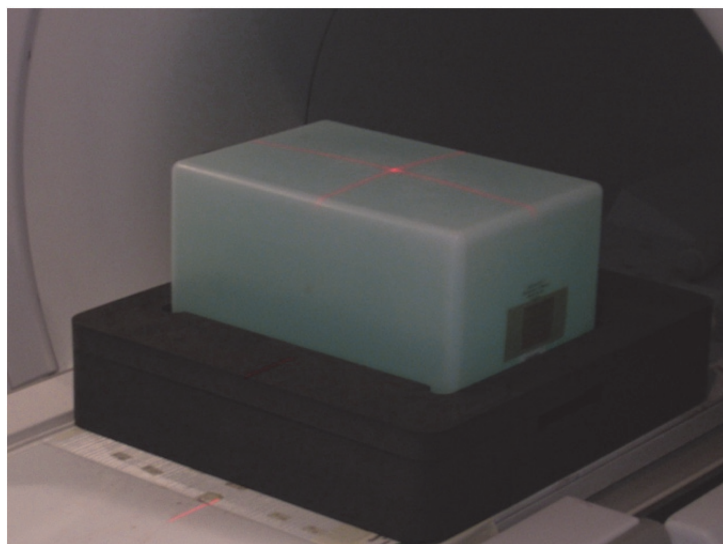


Illustration 9: MCQA SETUP WITH UNIFIED TL BRICK

4. Turn alignment button on.

Move table into bore until landmark lasers intersect with crosshairs on the coil as shown in Illustration 8.

Press 'Landmark' button

Place phantom on the coil. Ensure that the phantom is contained within the phantom positioner as shown in Illustration 9.

Press 'Advance to Scan' button.

3.3 MCQA Test

- 3.3.1 At the operator workstation, end the previous exam (if required).
- 3.3.2 From the Common Service Desktop, select the [Image Quality] tab, choose "Multi-Coil QA Tool", and then "Click here to start this tool" as shown in Illustration 10. Select 3.0T_DVw_32Ch_Cardiac_by_Invivo_Anterior_only.Unified.mctcfg as shown in Illustration 11.

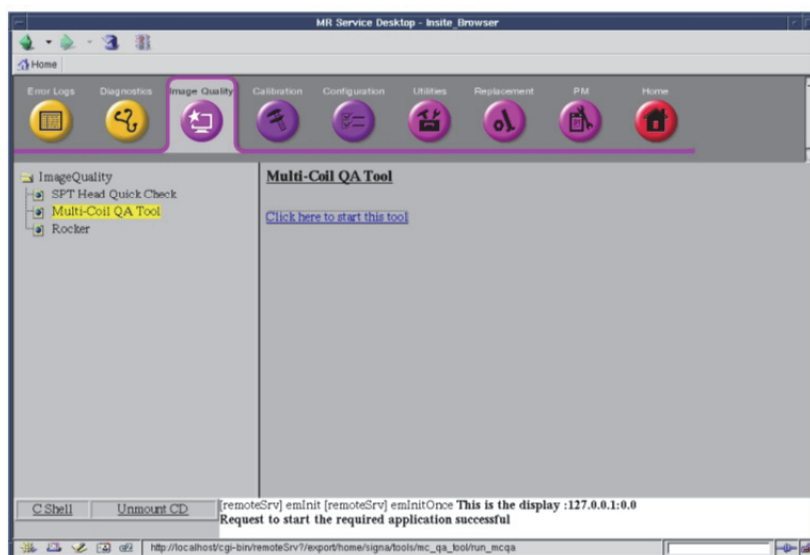


Illustration 10: MCQA from the Service Desktop

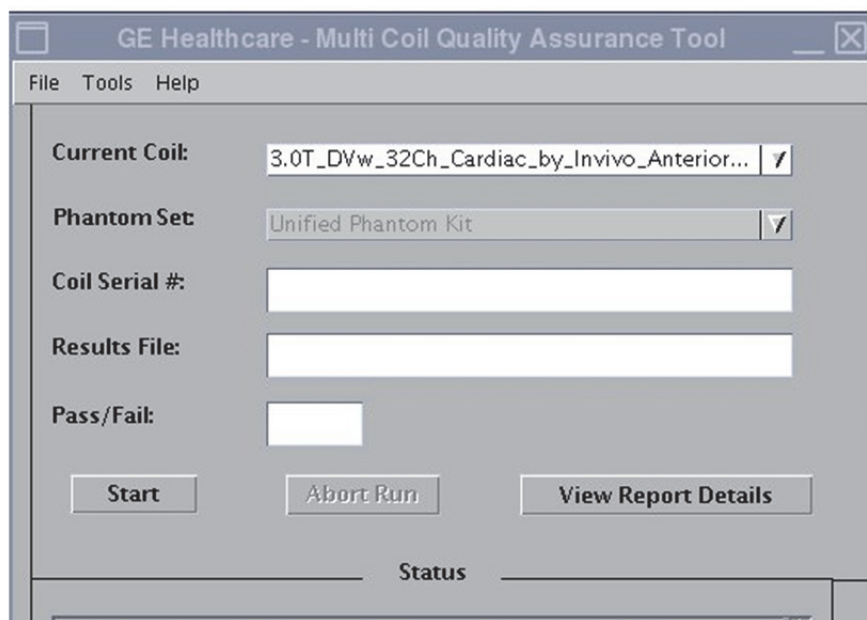


Illustration 11

3.3.3 Further instructions on using the MCQA tool can be found on the Service Methods CD at: Troubleshooting -> System -> Multi-Coil Quality Assurance Tool.

For troubleshooting: The relationship between MCQA image order, coil Receive Pin and coil element are the same (IMG 1 = Rx Pin 1 = Element 1). Illustration 12 below shows the element layout.

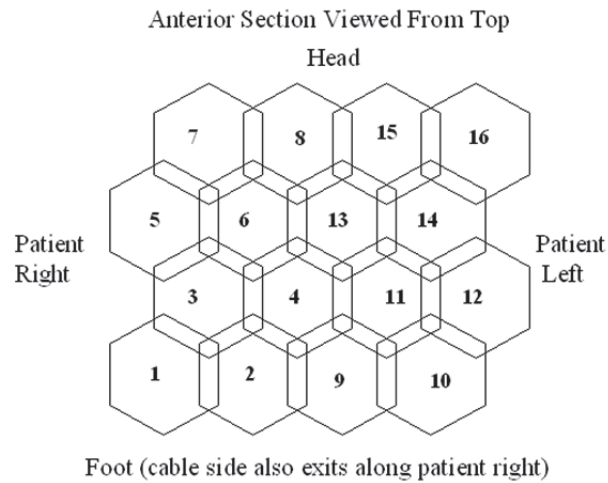


Illustration 12: Element layout for GEM Small Anterior Array

3.4 Periodic Quality Assurance Check

QA checks should be performed during installation and during planned maintenance. Perform the quality assurance checks as outlined below to ensure the coil is operating properly.

3.4.1 Check external cable for cracks or cuts.

3.4.2 Perform Section 3: 3.0T GEM SMALL ANTERIOR ARRAY MCQA TEST

SECTION 4: TROUBLESHOOTING

Note: The Invivo Anterior Array is not to be opened for any reason!

4.1 Mechanical Hardware Check

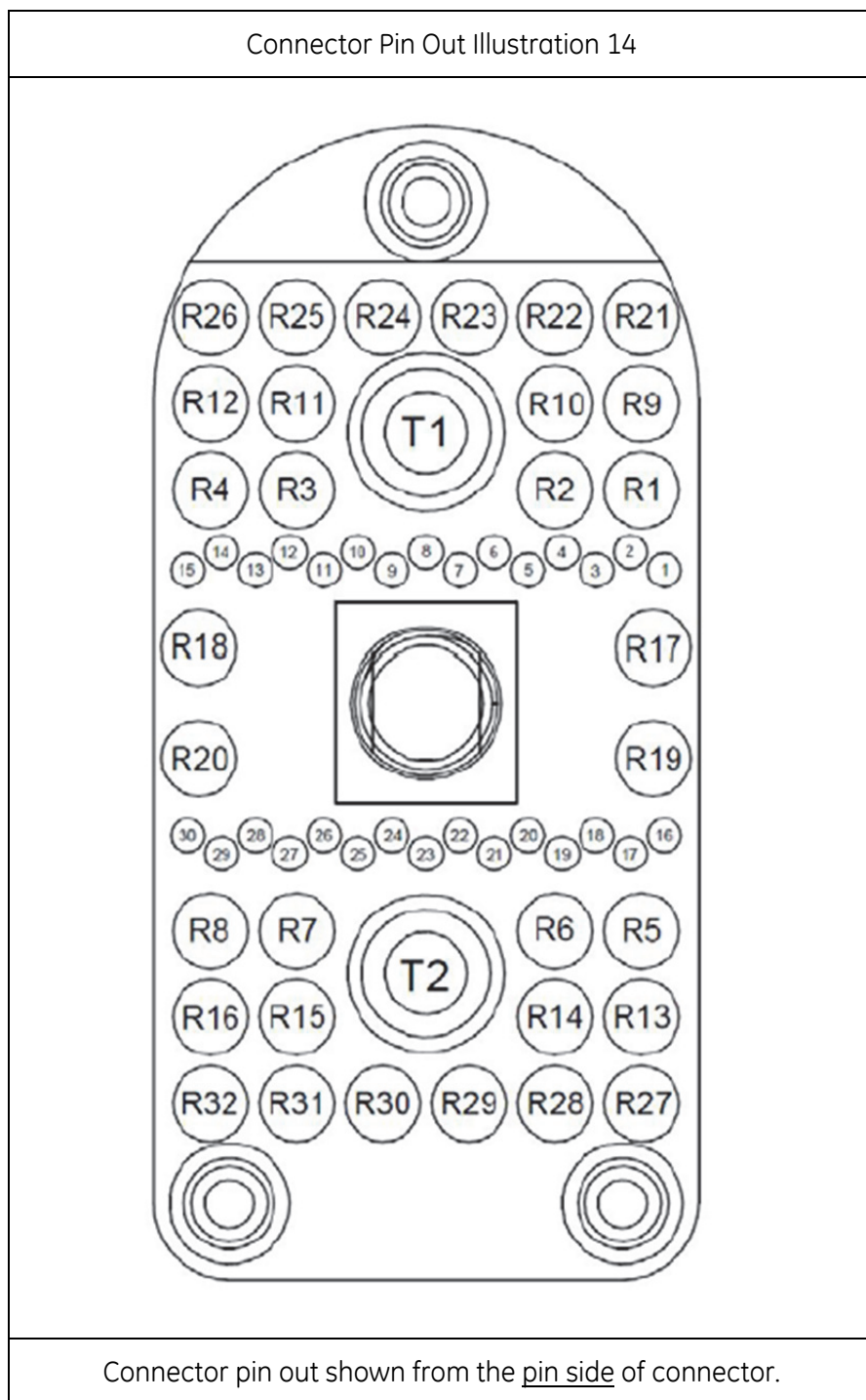
- Verify that there are no loops in the cables.
- Verify that there are no metal or ferromagnetic objects close to the coil, patient or magnet (i.e., safety pin, hair pin).
- Verify that the coil is properly positioned.

4.2 Electrical Hardware check

Visually inspect all coaxial connectors and pins on the system plug. If there are any broken, deformed or recessed pins or coaxial connectors, replace the coil component.

Coil Connector Illustration 13





NOTE: The center pins of the System Connector *receive coaxial connectors* are extremely fragile [Connector Pin Out Illustration , R1-R8]. **Do not attempt to make any measurements at these center pins.**

For GND measurements, use any of the receive coaxial pins' outer shields (R1-R8).

RESISTANCE MEASUREMENTS – TABLE 4-2-1

System Connector		Reading
Positive Lead	Negative Lead	
1	2	< 8Ω
3	4	< 8Ω
5	GND	Open
6	GND	Open
20	GND	140 ±40Ω
21	GND	< 5Ω
22	GND	140 ±40Ω
23	GND	< 8Ω

If one or more of the readings are out of specification, replace the coil. If all of the above readings are correct, redo Section 3: GEM SMALL ANTERIOR ARRAY MCQA test. If the problem still exists, replace the 3.0T Anterior Array.

4.3 Coil ID Troubleshooting Tips

If the following error is displayed, there may be a problem with the Coil ID coil or system hardware:
The prescribed RF coil and detected RF coil do not match.

Troubleshooting instructions are available through the Service Methods DVD via the path:
Troubleshooting -> Diagnostics -> Hardware Location -> Magnet Room -> Coil ID Functional Diagnostic

The diagnostic will display the contents of the coil ID chip whenever the connector is inserted. Follow the troubleshooting instructions if the correct ID is not displayed. When the troubleshooting is complete, and if the correct ID is still not displayed, replace the coil.

Troubleshooting example:

Coil ID diagnostic started

Reading Coil ID's...

Port 1 (P1) Connected

Port 3 (P3) Connected

Port 3 (P3): 1 Coil Detected:

--> Coil Name: GEM PA

Coil is VALID

Coil ID: GEMsiggjNFhn3pJ6xvsAqJoBQvFU0qb

Serial Number: 5c000000117e4e0f

Field Strength: 30000

Body Transmit: ENABLED

Port 1 (P1): 1 Coil Detected:

--> Coil Name: GEM Small AA

Coil is VALID

Coil ID: INVlppAbXy7sgvQeY29ZizFcq18H2Cb7

Serial Number: 430000001189dd0f

Field Strength: 30000

Body Transmit: ENABLED

SECTION 5 – MAINTENANCE

5.1 Coil Care

WARNING!

Detach coil connector from scanner before attempting to clean. Do not reattach after cleaning until coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.

CAUTION

Do not spray or pour cleaning solution directly on coil. Do not submerge coil in solution. The coil contains sensitive electronics components that could be damaged by the solution.

The 3.0T 32-Channel Cardiac Coil, posterior pads must be cleaned and stored using the following procedure:

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
- **Do not pour any cleaning solution directly on the coil!**
- Let the coil housing and pads dry before use.
- **Under no circumstances should the coil be placed into any type of sterilizer.**

SECTION 6 – RENEWAL PARTS

6.1 Field Replaceable Units

FIELD REPLACEABLE UNITS LIST – TABLE 6-1

Description	GE Part #	Invivo #
3.0T 32Ch Cardiac Coil - Anterior	5419534-2	4535-303-14471
GEM Small Anterior Array Phantom Positioner (2 pieces)	5406404-11	4535-303-03601

6.2 Customer Consumable Components

CUSTOMER CONSUMABLE COMPONENTS – TABLE 6-2

Description	GE Part #	Invivo #
Anterior Pad, 32ch Cardiac coil anterior only	5406404-10	4535-303-03611
Strap Set, 32ch Cardiac coil anterior only	5406404-12	4535-303-03621

Note: Pads and straps are designed as customer consumable components. They are not intended to be covered by any GE service contract. If pads and straps are replaced they should be billed to the customer.

SECTION 7 – UPGRADING TO GEM SMALL ANTERIOR ARRAY

7.1 How to upgrade

It is possible to upgrade a 3.0T 32Ch Cardiac Coil for MR750w Non-GEM only to a 3.0T GEM Small Anterior Array. To do so, order the following parts and bill the customer accordingly.

(Note: 32Ch Cardiac for MR750 cannot be upgraded to 3.0T GEM Small Anterior Array)

Description	GE Part #	Invivo #
GEM Small Anterior Array Phantom Positioner (2 pieces)	5406404-11	4535-303-03601
Strap Set, 32ch Cardiac coil anterior only	5406404-12	4535-303-03621

Revision History

Rev	Date	Author	Primary Reason for Change	DCN
1	8-16-11	R. Law	Issued	10049